Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 119501 | *Subject: Machine Learning* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option for the following questions (Answer any seven):*** **[2 x 7 = 14]**

(a) Bias-Variance trade-off mainly affects
(i) Hardware performance (iii) Model generalization
(ii) Internet speed (iv) Storage capacity

(b) Regression analysis is mainly used for
(i) Classification problems (iii) Predicting continuous values
(ii) Clustering data (iv) Image compression

(c) NumPy library in Python is mainly used for
(i) Game design (iii) Networking
(ii) Web development (iv) Numerical computation

(d) K-means algorithm is mainly used for
(i) Classification (iii) Clustering
(ii) Regression (iv) Reinforcement learning

(e) Activation functions are used to
(i) Introduce non-linearity (iii) Compress images
(ii) Increase storage (iv) Perform encryption

(f) Q-learning belongs to
(i) Supervised learning (iii) Reinforcement learning
(ii) Clustering (iv) Regression

(g) Random Forest is an ensemble of
(i) Neural networks (iii) Decision trees
(ii) Regression models (iv) K-means clusters

(h) Bagging is mainly used to
(i) Reduce variance (iii) Increase overfitting
(ii) Remove features (iv) Compress data

(i) Naive Bayes algorithm is based on
(i) Probability theory (iii) Neural networks
(ii) Clustering (iv) Sorting

(j) PCA is mainly used for
(i) Data encryption (iii) Sorting data
(ii) Dimensionality reduction (iv) Data transmission

Q.2 (a) Describe supervised, unsupervised and reinforcement learning with suitable practical examples. **[7]**

(b) What do you mean by Bias and variance? Discuss how high bias and high variance affect model performance in predictive analytics. **[7]**

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Q.3	(a)	Define lists, tuples and dictionaries in Python with suitable examples related to Machine Learning applications.	[7]
	(b)	Differentiate between clustering and classification with suitable examples from real-world applications.	[7]
Q.4	(a)	A healthcare system wants to classify patients as diabetic or non-diabetic based on medical records. Discuss which supervised learning algorithm would be suitable and justify your answer.	[7]
	(b)	Support Vector Machine uses soft margin and hard margin techniques for classification. Discuss their differences and practical applications.	[7]
Q.5	(a)	Differentiate between bagging and boosting with suitable examples used in Machine Learning applications.	[7]
	(b)	What are the limitations of Random Forest techniques? A Random Forest model contains 120 decision trees where each tree gives one prediction. If 84 trees vote for Class A and 36 trees vote for Class B, calculate the final predicted class and voting percentage for Class A.	[7]
Q.6	(a)	Differentiate between Q-learning and SARSA algorithms used in Reinforcement Learning with suitable examples.	[7]
	(b)	PCA is widely used for dimensionality reduction in Machine Learning. Discuss its advantages, limitations and practical applications in data preprocessing.	[7]
Q.7	(a)	Discuss single-layer perceptron and multilayer feed-forward neural networks with suitable examples. https://www.akubihar.com	[7]
	(b)	Activation functions and learning rules play an important role in Neural Networks. Discuss their significance in training deep learning models.	[7]
Q.8	(a)	A company wants to predict house prices using previous sales data. Discuss why regression analysis is suitable for this problem and mention practical limitations.	[7]
	(b)	A student wants to use TensorFlow and Keras for image classification projects. Discuss the advantages of these libraries in deep learning applications.	[7]

Q.9 Write Short notes on ***any two*** of the following: **[2 x 7 = 14]**

(a) Curse of Dimensionality
(b) Back-propagation Algorithm
(c) KNN
(d) GMM & EM Algorithm

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